

Synergy Level 3: Business

visual context

	Corton	Denham
Forecast earnings per share	\$5.00	\$1.50
Forecast dividend per share	\$3.00	\$0.80
Number of shares	1,000,000	600,000
Stock price	\$90	\$20

textual context

.As financial manager of Corton Inc., you are investigating a possible acquisition of Denham. You have the basic data given in the following table: Corton: Forecast earnings per share = \$5.00, Forecast dividend per share = \$3.00, Number of shares = 1,000,000, Stock price = \$90. Denham: Forecast earnings per share = \$1.50, Forecast dividend per share = \$0.80, Number of shares = 600,000, Stock price = \$20. You estimate that investors expect a steady growth of about 6% in Denham's earnings and dividends. Under new management, this growth rate would be increased to 8% per year without the need for additional capital.

Question:

How would the cost of the share offer change if the expected growth rate was not changed by the merger?

Options:

A: \$3,000,000; B: \$4,000,000; C: \$4,500,000 D: \$5,000,000

o3: Step 1: Value Denham as a stand-alone firm
With no change in Denham's growth rate, its price stays at the market price that reflects a 6 % growth rate:
 $P_{\text{Denham}} = \$20$ per share. Total market value of Denham $V_{\text{Denham}} = 600\,000 \text{ shares} \times \$20 = \$12\,000\,000$.
Step 2: Exchange ratio to be used
From the information given earlier in the problem (the stock-for-stock offer that Corton intends to use), Denham's shareholders are to receive 0.30 Corton share for each Denham share (given in the earlier part of the problem) so that the number of new Corton shares that must be issued is
 $N_{\text{new}} = 0.30 \times 600\,000 = 180\,000 \text{ shares}$.
Step 3: Cost of the share offer if Denham's growth is NOT increased
Corton's shares are trading at \$90, therefore the value of the shares that Corton will hand over is
 $\text{Cost} = N_{\text{new}} \times P_{\text{Corton}} = 180\,000 \times \$90 = \$16\,200\,000$.
Step 4: Incremental cost to Corton's shareholders
The real "cost" of the acquisition is what Corton gives up over and above the stand-alone value of Denham:
 $\text{Incremental cost} = \$16\,200\,000 - \$12\,000\,000 = \$4\,200\,000 \approx \$4\,000\,000$.
Thus, if the merger did NOT succeed in lifting Denham's growth rate, the stock offer would cost Corton about \$4 million.
\boxed{B}

Ground Truth: D: \$5,000,000